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ORIGINAL ARTICLE

Adaptive Online Learning and Robust 3-D Shape Servoing of Continuum and Soft Robots in Unstructured Environments

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Abstract

In this article, we present a novel and generic data-driven method to servo-control the 3-D shape of continuum and soft robots based on proprioceptive sensing feedback. Developments of 3-D shape perception and control technologies are crucial for continuum and soft robots to perform tasks autonomously in surgical interventions. However, owing to the nonlinear properties of continuum robots, one main difficulty lies in the modeling of them, especially for soft robots with variable stiffness. To address this problem, we propose a versatile learning-based adaptive shape controller by leveraging proprioception of 3-D configuration from fiber Bragg grating (FBG) sensors, which can online estimate the unknown model of continuum robot against unexpected disturbances and exhibit an adaptive behavior to the unmodeled system without priori data exploration. Based on a new composite adaptation algorithm, the asymptotic convergences of the closed-loop system with learning parameters have been proven by Lyapunov theory. To validate the proposed method, we present a comprehensive experimental study using two continuum and soft robots both integrated with multicore FBGs, including a robotic-assisted colonoscope and multisection extensible soft manipulators. The results demonstrate the feasibility, adaptability, and superiority of our controller in various unstructured environments, as well as phantom experiments.

Keywords: shape control, learning-based adaptive control, continuum and soft robots, fiber Bragg grating sensors

Introduction

Continuum and soft robots are composed of compliant materials and flexible structures, endowing them with high dexterity and compliance for a wide variety of application scenarios, such as industrial fields, human–robot interactions, and medical interventions. In view of the advantages of continuum and soft robots in terms of manipulating accessibility, intraoperative maneuverability, and inherent safety, they possess great potentials to be deployed in the treatment of various types of minimally invasive surgery (MIS).[1,2] However, the interference of nonlinear structural characteristics and unknown environmental disturbances would negatively affect the priori modeling of

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continuum and soft robots, hence accurate 3-D shape sensing and robust control are considered as major challenges for their efficient and safe implementations in unstructured scenarios as illustrated in Figure 1, especially for soft robots integrated with soft materials and structures that exhibit heterogeneous properties and variable stiffness.[3–5]

Considering that continuum and soft robot has an infinite number of degrees-of-freedom (DOFs) in configuration space,[2,4,6] real-time perception of its 3-D shape is essential to advance control strategies. To date, numerous types of sensing modalities have been exploited for shape sensing and position estimation of continuum robots in MIS, such as mechanics-based methods, intraoperative imaging modalities, magnetic resonance imaging, and electromagnetic tracking.[3] However, these methods suffer from their limitations concerning acquisition frequency, sensing accuracy, application versatility, and clinical compatibility.

Moreover, a variety of proprioceptive sensing methods have been developed for soft robotics to be more compact and efficient. Scharff et al.[6,7] proposed optical-based and color-based sensing networks for 3-D deformation reconstruction of pneumatic soft robots and developed machine learning algorithms for their shape parameterization. Strain gauges were utilized and embedded into the soft continuum manipulator for its shape reconstruction under a learning-based curve fitting approach.[8] Loo et al.[9] proposed a multimodal sensing scheme for state estimation of a multifinger soft gripper by incorporating recurrent neural networks (NNs) and an adaptive unscented Kalman filter (KF), which can not only handle system drift and hysteresis issues but also can eliminate the effects of sensor faults, as well as out-of-distribution sensor dynamics.

Recently, fiber Bragg grating (FBG) sensors have shown great potential in shape sensing and proprioception of continuum and soft robots due to their lightweight, high compatibility, and high feedback frequency without requiring line of sight.[10–12] Considering severe signal noises of FBG measurements in unstructured environments, the accumulative error along the longitudinal direction could defect the accuracy of the sensing result. Aiming to solve this problem, some sensor fusion using KFs[13–15] and learning-based approaches[16–18] have been developed. A model-based filtering algorithm for 3-D shape sensing of flexible endoscope using multicore FBGs was recently reported to be accurate and robust against the sensory noises and unexpected perturbations.[19]

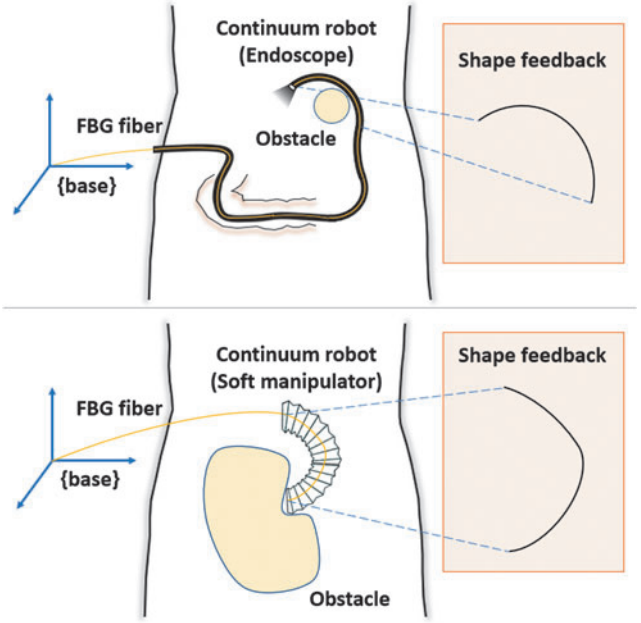


FIG. 1. Conceptual representation of 3-D shape sensing and control for continuum and soft robots, for example, flexible endoscope (*top*), and soft manipulator (*bottom*), in unstructured environments.

Using task-space position/shape with encoders information as feedback, various control methodologies of continuum and soft robots have been proposed for distal endpoint positioning and shape servoing.[20–22] According to modeling approaches, control strategies can be mainly classified into model-based and model-free methods. Model-based methods require *a priori* accurate modeling of the continuum robot, including kinematics/statics and dynamics models[23] based on certain assumptions and theories, for example, constant curvature,[1,24] beam theory,[25] and Cosserat rod theory.[26,27] In model-based methods, the modeling of continuum and soft robots should be identified before implementation which would result in unknown deviations in the presence of backlash, friction, and hysteresis. Besides, their strong assumptions might not be validated when with unexpected disturbances.

In addition to sequential actuated approaches, soft robots actuated by pneumatic,[6] fluidic,[28,29] and dielectric elastomer (DE)[30] have been widely investigated as well, and their deformation modeling together with shape control still remains as challenges.[22,31] Nevertheless, energy-shaping control of a pneumatic soft robot[32] has emerged as a promising trend to compensate for disturbances with modeling from a port-Hamiltonian approach. Besides, some state-of-the-art works have also explored external force estimation and regulation tasks of soft robots.[33]

Model-free control strategies do not require *a priori* knowledge about the model of continuum and soft robot and its modeling uncertainties, which are instead acquired by estimation methods. They are applicable to the highly nonlinear or nonuniform systems that can be affected by unknown perturbations,[4] making priori models invalid. Yip and Camarillo[34,35] proposed optimal control methods with optimization-based online estimations of the kinematics Jacobian and stiffness of a continuum robot. Alambeigi et al.[14,21] presented a model-independent KF to estimate the deformation of an unmodeled continuum robot by fusing eye-to-hand camera vision and FBG signals; lately, they introduced an optimization-based control framework for visual servoing of the robot with online model update using Broyden's method. Broyden's update algorithm was also applied to evaluate derivatives on the deformable membrane for optimization-based control of a soft robotic mannequin.[36]

An adaptive KF was adopted for a soft continuum robot driven by pneumatic artificial muscles (PAMs) to update its Jacobian matrix,[37] whose entries were assumed to vary linearly. Bruder et al.[38] introduced a data-driven model predictive control (MPC) strategy for a PAM-based soft robot by incorporating a Koopman-based nonlinear system identification procedure that can capture the inherent input-output behavior. Moreover, Fang et al.[31] developed a generic geometry-based framework for kinematics modeling of soft robots, whose bodies and different types of actuators can be

formulated as the geometric elements, and they accordingly proposed an image-based calibration procedure to update the model parameters.

With the advancements of artificial intelligence, control strategies using machine learning, which can identify and learn the model/motion-based policies of continuum and soft robots, have gained great attention.[5,39] Among them, NNs were commonly utilized for modeling of continuum and soft robots (e.g., inverse kinematics).[8,11,40–43] Gaussian process regression has been also investigated as an alternative approach to approximate models of soft robots.[44,45] Thuruthel et al.[46] implemented reinforcement learning (RL) for soft robot control through learning control actions from recurrent NN-based dynamics model. Model-free RL algorithms are also developed to control soft robot using simulation data instead of analytical modeling.[47,48]

Although these approaches alleviate the need for accurate modeling of continuum and soft robots, large amounts of training data and offline identification are heavily required, hampering their practical deployments.[5] Due to the probabilistic properties of these learning-based approaches, another major concern is the difficulty of deriving stability conditions and even asymptotic convergences of control error and learning parameters,[49] which may raise safety problems in unstructured contexts.

To guarantee convergent performance when using learning-based methods, Braganza et al.[50] utilized a NN to feedforward account for the nonlinear dynamics uncertainty in a soft continuum robot, while maintaining the boundedness of NN output together with the tracking error convergence under a nonlinear feedback controller. Tang et al.[49] developed an integration of iterative learning and MPC for soft robots that can online update the model parameters and guarantee asymptotic stability. However, control strategies of continuum and soft robots with stability analysis concerning the modeling error caused by online learning have not been reported.[51] Nevertheless, Melingui et al.[52] addressed the stability of an adaptive NN controller that learns the kinematics of continuum robots; training an emulator is necessarily required before the control task.

Rigorous analysis in the presence of NN approximation error has not been considered as well.[53,54] Moreover, although some existing methodologies have generic control formulations, they were only validated with motions in 2-D space by adding in-plane perturbations, while neglecting those out-of-plane ones and, thus, without investigations in more complex shapes.[55–57] Overall, there are open problems in 3-D shape control of continuum and soft robots enabling them to perform complex tasks in sensitive environments,[4] especially for learning-based approaches, which are expected to integrate the advantages emphasized on: no priori data requirement, stability with convergences of control error and learning parameters, online error compensation, and accurate shape feedback.[2,5]

Toward these limitations, this article proposes a robust data-driven control framework that can automatically regulate the 3-D shape of FBG-equipped proprioceptive continuum and soft robot in unstructured environments. To the best of the authors' knowledge, this is the first work in control with guaranteed stability for continuum and soft robots using 3-D shape feedback (e.g., FBG sensing), based on the development of a novel NN-based adaptive control strategy for online learning the robot's differential kinematics and simultaneous shape servoing. The proposed algorithm does not require priori data exploration and can concurrently guarantee the convergent performance of control error and learning parameters and online error compensation with unknown external conditions compared with the state-of-the-art methods.[5] The main contributions are summarized as follows:

1. Development of a data-driven adaptive controller by incorporating FBG-based proprioception of 3-D configuration and a learning-based shape flow predictor, to online update the unknown differential kinematics of the continuum robot without its priori knowledge and to simultaneously servo-control its 3-D shape.
2. Consideration of the online modeling (parameters) error, NN approximation error, and unknown environmental conditions in development of learning-based control for continuum robot. Under the proposed controller together with a novel composite adaptation algorithm jointly introducing the shape estimation and flow errors of the robot, the asymptotic stability of the closed-loop system and the convergences of the shape control and estimation errors to zeros are proved using Lyapunov method, thus showing the convergence of learning parameters.
3. Design and implementation of a robust and accurate filtering algorithm for 3-D shape sensing of the continuum robot by solely embedding a multicore FBG fiber, which is independent of external sensors and maintains high sensing quality against disturbances. More configuration-space features computed from the distributed FBGs can be leveraged for online learning and shape control than only endpoint position.

The proposed control framework is generic and efficient that can be implemented on different kinds of continuum and soft robots,[23,29,57] as well as when using other external or proprioceptive sensing modalities as discussed above. We have validated the controller using two FBG-equipped proprioceptive continuum robots, including a robotic-assisted colonoscope system and multisection extensible soft manipulators. The results demonstrate the versatility and adaptability of our algorithm in various unstructured environments, as well as two phantom experiments for colonoscopy and oropharyngeal (OP) sampling.

Mathematical Modeling

Problem statement

In this work, we can trace the Cartesian positions of the points on the continuum robot from an FBG-based proprioception system with the compact integration of FBG fiber inside a channel through the entire robot. The 3-D shape feature is constructed from the selected points' positions and utilized as the feedback signal for online model learning and simultaneous shape servoing. Given a desired shape of the continuum robot, the objective is to design a velocity controller on the actuation input such that its shape asymptotically approaches to the desired one without any priori and explicit identification of the robot's model. To address this problem, we first derive the following differential kinematics of continuum robots and shape feature Jacobian matrix and then design a data-driven adaptive control algorithm with rigorous stability analysis in the next section.

Kinematics of continuum robots

To represent the state of the continuum robot in its configuration space, we first denote l points selected on the robot with their 3-D Cartesian positions $\boldsymbol{r}_i \in \mathbb{R}^3, i \in \{1, 2, \ldots, l\}$, which are measured from the FBG sensors, and the actuator position input of the robot $\boldsymbol{q} = [\, q_1 \quad q_1 \quad \cdots \quad q_n \,]^\mathsf{T} \in \mathbb{R}^n$. Then, we define a vector for the Cartesian positions of l selected points $\boldsymbol{r} \in \mathbb{R}^{3l}$ as

$$\boldsymbol{r} = \begin{bmatrix} \boldsymbol{r}_1^\mathsf{T} & \boldsymbol{r}_2^\mathsf{T} & \ldots & \boldsymbol{r}_l^\mathsf{T} \end{bmatrix}^\mathsf{T} \tag{1}$$

The kinematics of the continuum robot mapping the selected points on the robot $\boldsymbol{r}$ to the actuation position $\boldsymbol{q}$ can be expressed by a smooth nonlinear unknown function as $\boldsymbol{r} = \boldsymbol{r}(\boldsymbol{q}) : \mathbb{R}^n \mapsto \mathbb{R}^{3l}$, the differential formulation of which as the differential kinematic equation yields

$$\dot{\boldsymbol{r}} = \boldsymbol{J}_r(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{2}$$

where $\boldsymbol{J}_r(\boldsymbol{q}) \in \mathbb{R}^{3l \times n}$ represents the unknown Jacobian matrix of the continuum robot. Note that the actuator velocity input of the continuum robot $\dot{\boldsymbol{q}}$ can be exactly set from the proposed controller, and its acceleration $\ddot{\boldsymbol{q}}$ is bounded.[58]

Shape feature Jacobian matrix

Using the selected points $\boldsymbol{r}_i,\ i \in \{1, 2, \ldots, l\}$ to describe the overall 3-D shape of the continuum robot, we construct a vector of shape feature as $\boldsymbol{x} = [\, x_1 \quad x_2 \quad \ldots \quad x_m \,]^\mathsf{T} \in \mathbb{R}^m$ with a smooth function $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{r}) : \mathbb{R}^{3l} \mapsto \mathbb{R}^m$, which will be explicitly defined afterward. The selected points' velocity $\dot{\boldsymbol{r}}$ is related to the shape feature flow $\dot{\boldsymbol{x}} \in \mathbb{R}^m$ as

$$\dot{\boldsymbol{x}} = \boldsymbol{J}_x(\boldsymbol{r})\dot{\boldsymbol{r}} \tag{3}$$

where $\boldsymbol{J}_x(\boldsymbol{r}) \in \mathbb{R}^{m \times 3l}$ is the shape feature Jacobian matrix. In a combination of Equations (2) and (3), we can relate the actuation velocity $\dot{\boldsymbol{q}}$ to the shape flow $\dot{\boldsymbol{x}}$ as

$$\dot{\boldsymbol{x}} = \underbrace{\boldsymbol{J}_x(\boldsymbol{r})\boldsymbol{J}_r(\boldsymbol{q})}_{\boldsymbol{J}_c(\boldsymbol{q})}\dot{\boldsymbol{q}} \tag{4}$$

where $\boldsymbol{J}_c(\boldsymbol{q}) = \boldsymbol{J}_x(\boldsymbol{r})\boldsymbol{J}_r(\boldsymbol{q}) \in \mathbb{R}^{m \times n}$ is denoted as the combined deformation Jacobian matrix of the robot mapping its actuator velocity $\dot{\boldsymbol{q}}$ to the shape flow feedback $\dot{\boldsymbol{x}}$. Note that Equation (4) can be rewritten as

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_m \end{bmatrix} = \begin{bmatrix} \boldsymbol{J}_{c,1}(\boldsymbol{q}) \\ \vdots \\ \boldsymbol{J}_{c,m}(\boldsymbol{q}) \end{bmatrix} \dot{\boldsymbol{q}} \tag{5}$$

where $\boldsymbol{J}_{c,i}(\boldsymbol{q}) \in \mathbb{R}^{1 \times n},\ i \in \{1, 2, \ldots, m\}$ represents the i-th row of $\boldsymbol{J}_c(\boldsymbol{q})$.

NN model

Considering that the combined Jacobian matrix $\boldsymbol{J}_c(\boldsymbol{q})$ has unknown and time-varying elements, $\boldsymbol{J}_c(\boldsymbol{q})$ is hard to be directly estimated using a conventional adaptive mechanism. For each row of $\boldsymbol{J}_c(\boldsymbol{q})$ in the model (5), it can be approximated by a NN with the actuation $\boldsymbol{q}$ being the NN input, represented by

$$\boldsymbol{J}_{ci}(\boldsymbol{q}) = (\boldsymbol{W}_i\theta_i(\boldsymbol{q}))^\mathsf{T},\ i \in \{1, 2, \ldots, m\} \tag{6}$$

where $\boldsymbol{W}_i \in \mathbb{R}^{n \times k},\ i \in \{1, 2, \ldots, m\}$, represents the ideal weight matrix of the i-th NN with k neurons, and $\boldsymbol{\theta}_i(\boldsymbol{q}) \in \mathbb{R}^k,\ i \in \{1, 2, \ldots, m\}$, is the corresponding vector of the activation functions. The j-th neuron, $\theta_{ij}(\boldsymbol{q}) : \mathbb{R}^n \mapsto \mathbb{R}$, $j \in \{1, 2, \ldots, k\}$, of the activation functions $\boldsymbol{\theta}_i(\boldsymbol{q})$ is defined using the radial basis function (RBF)[53,59] as

$$\theta_{ij}(\boldsymbol{q}) = \boldsymbol{e}^{\frac{-\|\boldsymbol{q} - \mu_{ij}\|^2}{\sigma_{ij}^2}} \tag{7}$$

where $\mu_{ij} \in \mathbb{R}^n$ is a vector representing the center value of the j-th neuron, and σ_{ij} is the corresponding width value.

Data-Driven Control Algorithm

A novel data-driven control strategy using proprioceptive sensing feedback and adaptive NN online learning, which is robust to system uncertainty and external disturbances, is designed for 3-D shape servoing of the continuum robot.

Model modifications and NN approximation

In consideration of the system uncertainties and external disturbances in unstructured environments, we take an unknown and time-varying disturbance term $\boldsymbol{d} \in \mathbb{R}^m$ into Equation (4) as

$$\dot{\boldsymbol{x}} = \boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}} + \boldsymbol{d} \tag{8}$$

Note that the disturbance $\boldsymbol{d}$ and its time derivative $\dot{\boldsymbol{d}}$ are bounded, and $\boldsymbol{d} \to \boldsymbol{0}$ as $\dot{\boldsymbol{q}} \to \boldsymbol{0}$.

To approximate the combined Jacobian matrix $\boldsymbol{J}_c(\boldsymbol{q})$, we adopt m adaptive NNs $\boldsymbol{W}_i\boldsymbol{\theta}_i(\boldsymbol{q}),\ i \in \{1, 2, \ldots, m\}$, for online learning of m rows of $\boldsymbol{J}_c(\boldsymbol{q})$, respectively, given by

$$\boldsymbol{J}_c(\boldsymbol{q}) = [\, \boldsymbol{W}_1\boldsymbol{\theta}_1(\boldsymbol{q}) \quad \boldsymbol{W}_2\boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \boldsymbol{W}_m\boldsymbol{\theta}_m(\boldsymbol{q}) \,]^\mathsf{T} + \boldsymbol{E}_J \tag{9}$$

where the basis of the activation functions $\boldsymbol{\theta}_i(\boldsymbol{q})$, $i \in \{1, 2, \ldots, m\}$, can be determined according to the rough mapping from joint space to configuration space of continuum robot, which will be detailed in Experimental Setup section. In Equation (9), $\boldsymbol{E}_J \in \mathbb{R}^{m \times n}$ denotes a matrix representing the functional approximation error of the NN learning,[53] which is upper bounded and could decrease with the increase in the number of neurons k.

Using NN approximation to the combined Jacobian matrix in Equation (9), the measurable shape flow $\dot{\boldsymbol{x}}$ using FBG sensing in Equation (8) can be further as

$$\dot{\boldsymbol{x}} = [\, \boldsymbol{W}_1\boldsymbol{\theta}_1(\boldsymbol{q}) \quad \boldsymbol{W}_2\boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \boldsymbol{W}_m\boldsymbol{\theta}_m(\boldsymbol{q}) \,]^\mathsf{T}\dot{\boldsymbol{q}} + \boldsymbol{\delta} \tag{10}$$

where $\boldsymbol{\delta} \in \mathbb{R}^m$ is a combined perturbation term that describes the modeling disturbances $\boldsymbol{d}$ together with NN approximation error $\boldsymbol{E}_J$ given by

$$\boldsymbol{\delta} = \boldsymbol{E}_J\dot{\boldsymbol{q}} + \boldsymbol{d} \tag{11}$$

Considering the boundedness of $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$, $\boldsymbol{d}$, $\dot{\boldsymbol{d}}$, $\boldsymbol{E}_J$, and its time derivative $\dot{\boldsymbol{E}}_J$,[53] the combined perturbation $\boldsymbol{\delta}$ and its time derivative $\dot{\boldsymbol{\delta}} \in \mathbb{R}^m$ are bounded from Equation (11) as

$$||\boldsymbol{\delta}|| \leq b_{\delta 1}, \quad \| \dot{\boldsymbol{\delta}} \| \leq b_{\delta 2} \tag{12}$$

where $b_{\delta 1}$ and $b_{\delta 2}$ are positive constants.

Since we do not have *a priori* knowledge of the deformation model of the continuum robot, as well as the ideal NN weight matrices (i.e., $\boldsymbol{J}_c(\boldsymbol{q})$ and $\boldsymbol{W}_i$, $i \in \{1, 2, \ldots, m\}$, are all unknown), we estimate the combined Jacobian matrix denoted by $\widehat{\boldsymbol{J}}_c(\boldsymbol{q}) \in \mathbb{R}^{m \times n}$ using m estimated NNs $\widehat{\boldsymbol{W}}_i \boldsymbol{\theta}_i(\boldsymbol{q}), i \in \{1, 2, \ldots, m\}$, as shown in Figure 2 and expressed by

$$\widehat{\boldsymbol{J}}_c(\boldsymbol{q}) = \left[\widehat{\boldsymbol{W}}_1 \boldsymbol{\theta}_1(\boldsymbol{q}) \quad \widehat{\boldsymbol{W}}_2 \boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \widehat{\boldsymbol{W}}_m \boldsymbol{\theta}_m(\boldsymbol{q}) \right]^{\mathsf{T}} \tag{13}$$

where $\widehat{\boldsymbol{W}}_i \in \mathbb{R}^{n \times k}$, $i \in \{1, 2, \ldots, m\}$, is the estimated NN weight matrix. To online tune the estimated NN weights and approximate the robot's model, a composite adaptation law for iterative evolution of the weights is proposed. The prediction error[51] (also called estimation error in this article) of the shape flow is minimized by a robust shape flow predictor, and two vectors of NN weights for linear parameterization of the ideal and estimated combined Jacobian matrices are defined, which are derived in detail as follows.

Learning-based shape flow predictor

To minimize the prediction error[51] of shape flow for learning the unknown differential kinematics of continuum robot in Equation (8), we iteratively predict the shape flow denoted by $\dot{\hat{\boldsymbol{x}}} \in \mathbb{R}^m$ and compare its difference with the measurable shape flow from FBG sensing results. For that, a robust learning-based shape flow predictor composed of the NN online approximation and sliding mode method[54] is derived as

$$\dot{\hat{\boldsymbol{x}}} = \Big[\underbrace{\widehat{\boldsymbol{W}}_1 \boldsymbol{\theta}_1(\boldsymbol{q}) \quad \widehat{\boldsymbol{W}}_2 \boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \widehat{\boldsymbol{W}}_m \boldsymbol{\theta}_m(\boldsymbol{q})}_{\widehat{\boldsymbol{J}}_c(\boldsymbol{q})} \Big]^{\mathsf{T}} \dot{\boldsymbol{q}} + \alpha_x \tilde{\boldsymbol{x}} + \beta_x sat(\tilde{\boldsymbol{x}}) \tag{14}$$

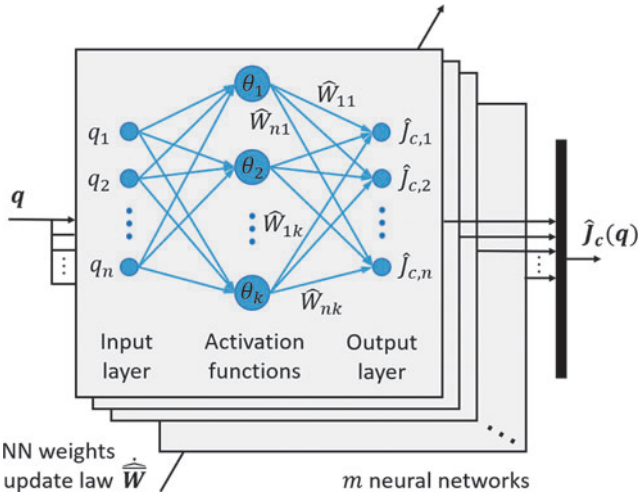


FIG. 2. RBF NNs for Jacobian rows approximation, where NN weights are updated by the adaptation algorithm. NNs, neural networks; RBF, radial basis function.

where α_x and β_x are positive constants; sat$(\cdot)$ is a continuous saturation function as the smoothed implementation of sliding mode method, which can robustly remedy the system disturbances $\boldsymbol{d}$ and NN approximation error $\boldsymbol{E}_J$, while avoiding chattering phenomenon[51]; and $\tilde{\boldsymbol{x}} \in \mathbb{R}^m$ represents the shape estimation error defined by

$$\tilde{\boldsymbol{x}} = \boldsymbol{x} - \hat{\boldsymbol{x}} \tag{15}$$

which is the difference between the measurable shape $\boldsymbol{x}$ regarded as the ground truth with the estimated shape $\hat{\boldsymbol{x}}$, that is, time integral of the predicted shape flow. The asymptotic convergence of this difference $\tilde{\boldsymbol{x}}$ to zero can be guaranteed in the stability analysis. Taking the time derivative of $\tilde{\boldsymbol{x}}$ in Equation (15), the prediction error of shape flow defined as shape flow error $\dot{\tilde{\boldsymbol{x}}} \in \mathbb{R}^m$ is derived from Equations (14) and (15) as

$$\begin{aligned} \dot{\tilde{\boldsymbol{x}}} &= \dot{\boldsymbol{x}} - \dot{\hat{\boldsymbol{x}}} \\ &= \dot{\boldsymbol{x}} - \widehat{\boldsymbol{J}}_c(\boldsymbol{q}) \dot{\boldsymbol{q}} - \alpha_x \tilde{\boldsymbol{x}} - \beta_x \text{sat}(\tilde{\boldsymbol{x}}) \end{aligned} \tag{16}$$

which is acquired by calculating the difference between the FBG measurement $\dot{\boldsymbol{x}}$ and the predictor $\dot{\hat{\boldsymbol{x}}}$ using NN-estimated Jacobian $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})$, actuation velocity $\dot{\boldsymbol{q}}$, together with the integral of the difference w.r.t. time, that is, the shape estimation error $\tilde{\boldsymbol{x}}$, whose initial value is set by $\tilde{\boldsymbol{x}} = \boldsymbol{x}$ at $t = 0$ because we do not have *a priori* knowledge of it.

Vectors of NN weight matrices

We linearly parameterize the combined Jacobian in terms of the NN weights to online update the unknown weights and learn the Jacobian for the controller. Let us group the columns of m ideal NN weight matrices $\mathbf{W}_i$ and m estimated ones $\widehat{\boldsymbol{W}}_i$, $i \in \{1, 2, \ldots, m\}$, respectively, into two vectors of them as $\overline{\boldsymbol{W}} \in \mathbb{R}^{kmn}$ and $\overline{\widehat{\boldsymbol{W}}} \in \mathbb{R}^{kmn}$ given by

$$\overline{\boldsymbol{W}} = \left[\boldsymbol{W}_{11}^{\mathsf{T}} \quad \boldsymbol{W}_{12}^{\mathsf{T}} \quad \ldots \boldsymbol{W}_{1k}^{\mathsf{T}} \quad \boldsymbol{W}_{21}^{\mathsf{T}} \quad \ldots \quad \boldsymbol{W}_{mk}^{\mathsf{T}} \right]^{\mathsf{T}} \tag{17}$$

$$\overline{\widehat{\boldsymbol{W}}} = \left[\widehat{\boldsymbol{W}}_{11}^{\mathsf{T}} \quad \widehat{\boldsymbol{W}}_{12}^{\mathsf{T}} \quad \cdots \widehat{\boldsymbol{W}}_{1k}^{\mathsf{T}} \quad \widehat{\boldsymbol{W}}_{21}^{\mathsf{T}} \quad \ldots \quad \widehat{\boldsymbol{W}}_{mk}^{\mathsf{T}} \right]^{\mathsf{T}} \tag{18}$$

where $\boldsymbol{W}_{ij}$ and $\widehat{\boldsymbol{W}}_{ij}$ are the j-th columns, $j \in \{1, 2, \ldots, k\}$, of the i-th ideal and estimated NN weight matrix $\boldsymbol{W}_i$ and $\widehat{\boldsymbol{W}}_i$, $i \in \{1, 2, \ldots, m\}$, respectively.

Composite adaptation for online weight estimation

Define two diagonal block matrices $\Theta(\boldsymbol{q}) \in \mathbb{R}^{km \times m}$ and $\boldsymbol{Q}(\dot{\boldsymbol{q}}) \in \mathbb{R}^{kmn \times km}$, respectively, by grouping the activation functions $\boldsymbol{\theta}_i(\boldsymbol{q})$, $i \in \{1, 2, \ldots, m\}$, and km actuator input $\dot{\boldsymbol{q}}$ as

$$\Theta(\boldsymbol{q}) = \text{diag}\Big(\underbrace{\boldsymbol{\theta}_1(\boldsymbol{q}), \ \boldsymbol{\theta}_2(\boldsymbol{q}), \ \cdots, \ \boldsymbol{\theta}_m(\boldsymbol{q})}_{\boldsymbol{\theta}_i(\boldsymbol{q}), \ i \in \{1, 2, \ldots, m\}} \Big) \tag{19}$$

$$\boldsymbol{Q}(\dot{\boldsymbol{q}}) = \text{diag}\Big(\underbrace{\dot{\boldsymbol{q}}, \ \dot{\boldsymbol{q}}, \ \cdots, \ \dot{\boldsymbol{q}}}_{km \ \dot{\boldsymbol{q}}} \Big) \tag{20}$$

The first term of the FBG-measured shape flow $\dot{\boldsymbol{x}}$ in Equation (10) approximated by the ideal NNs can be linearly parameterized using Equations (17), (19), and (20) as

$$\begin{aligned}&[\boldsymbol{W}_1\boldsymbol{\theta}_1(\boldsymbol{q}) \quad \boldsymbol{W}_2\boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \boldsymbol{W}_m\boldsymbol{\theta}_m(\boldsymbol{q})]^{\mathsf{T}}\dot{\boldsymbol{q}}\\&\quad=\Theta^{\mathsf{T}}(\boldsymbol{q})[\boldsymbol{W}_1 \quad \boldsymbol{W}_2 \quad \ldots \quad \boldsymbol{W}_m]^{\mathsf{T}}\dot{\boldsymbol{q}}\\&\quad=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\overline{\boldsymbol{W}}\end{aligned} \tag{21}$$

as well as the NN approximation of the predicted shape flow $\dot{\hat{\boldsymbol{x}}}$ in Equation (14) formulated as

$$\begin{aligned}&\left[\widehat{\boldsymbol{W}}_1\boldsymbol{\theta}_1(\boldsymbol{q}) \quad \widehat{\boldsymbol{W}}_2\boldsymbol{\theta}_2(\boldsymbol{q}) \quad \cdots \quad \widehat{\boldsymbol{W}}_m\boldsymbol{\theta}_m(\boldsymbol{q})\right]^{\mathsf{T}}\dot{\boldsymbol{q}}\\&\quad=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widehat{\overline{\boldsymbol{W}}}\end{aligned} \tag{22}$$

where $\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})$ is also equivalent to $\Theta^{\mathsf{T}}(\boldsymbol{q})\otimes\dot{\boldsymbol{q}}^{\mathsf{T}}$ using Kronecker Product.[14] Then the measurable and predicted shape flow $\dot{\boldsymbol{x}}$ and $\dot{\hat{\boldsymbol{x}}}$ can be expressed from the linearly parameterized formulations in Equations (21) and (22) as

$$\dot{\boldsymbol{x}}=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\overline{\boldsymbol{W}}+\boldsymbol{\delta} \tag{23}$$

$$\dot{\hat{\boldsymbol{x}}}=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widehat{\overline{\boldsymbol{W}}}+\alpha_x\tilde{\boldsymbol{x}}+\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}) \tag{24}$$

and the difference between the ideal and estimated NN approximations to the combined Jacobian matrices multiplying the actuation velocity $\dot{\boldsymbol{q}}$, that is, the difference between $\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}}$ and $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})\dot{\boldsymbol{q}}$ is calculated by

$$\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}}-\widehat{\boldsymbol{J}}_c(\boldsymbol{q})\dot{\boldsymbol{q}}=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widetilde{\overline{\boldsymbol{W}}}+\boldsymbol{E}_J\dot{\boldsymbol{q}} \tag{25}$$

where $\widetilde{\overline{\boldsymbol{W}}}=\overline{\boldsymbol{W}}-\widehat{\overline{\boldsymbol{W}}}\in\mathbb{R}^{kmn}$ denotes the estimation error of NN weights in vector form. From Equations (23) and (24), we can further derive the linearly parameterized shape flow error $\dot{\tilde{\boldsymbol{x}}}$ as

$$\dot{\tilde{\boldsymbol{x}}}=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widetilde{\overline{\boldsymbol{W}}}+\boldsymbol{\delta}-\alpha_x\tilde{\boldsymbol{x}}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}) \tag{26}$$

In addition to the above derivations for the adaptive update of the NN weights, we introduce the filtered shape estimation error $\boldsymbol{r}_x\in\mathbb{R}^m$ from Equation (26) defined by

$$\boldsymbol{r}_x=\dot{\tilde{\boldsymbol{x}}}+\alpha_x\tilde{\boldsymbol{x}}=\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widetilde{\overline{\boldsymbol{W}}}+\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}) \tag{27}$$

which can be minimized using the following adaptation law and is available to ensure the asymptotic regulation of the shape flow error $\dot{\tilde{\boldsymbol{x}}}$ compared with other forms difficult for this objective, for example, simply $\dot{\tilde{\boldsymbol{x}}}$.

Consequently, the vector of the estimated NN weights $\widehat{\overline{\boldsymbol{W}}}$ can be iteratively updated using a composite adaptation rule modified from Slotine-Li method[51] and proposed as follows:

$$\begin{aligned}\dot{\widehat{\overline{\boldsymbol{W}}}}=&\Gamma_W^{-1}\mathrm{proj}\{k_e\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{e}\\&+k_x\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\tilde{\boldsymbol{x}}+k_r\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{r}_x\}\end{aligned} \tag{28}$$

where k_e, k_x, and k_r are positive constants, Γ_W is a positive definite and diagonal gain matrix, $\boldsymbol{e}=\boldsymbol{x}-\boldsymbol{x}_d\in\mathbb{R}^m$ denotes the FBG-shape feedback error (control error), $\boldsymbol{x}_d\in\mathbb{R}^m$ represents the desired shape of the continuum robot, and $\mathrm{proj}\{\cdot\}$ is a projection operator to guarantee the estimated NN weights bounded.[60] In Equation (28), the first term of the shape control error is for the compensation of the difference between the estimated and real Jacobian matrices. The second term of the shape estimation error is to account for a cross term in the stability analysis. The third term about the filtered shape estimation error is for the online gradient descending minimization of the errors about the estimated model parameters.

Controller design and stability analysis

Constructing the estimated combined Jacobian $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})$ from the NNs online learning by Equation (13), the controller is designed as

$$\dot{\boldsymbol{q}}=-k_c\widehat{\boldsymbol{J}}_c^{+}(\boldsymbol{q})\boldsymbol{e}-k_s\widehat{\boldsymbol{J}}_c^{+}(\boldsymbol{q})\mathrm{sat}(\boldsymbol{e}) \tag{29}$$

where k_c and k_s are positive constants, $\widehat{\boldsymbol{J}}_c^{+}(\boldsymbol{q})\in\mathbb{R}^{n\times m}$ represents the Moore–Penrose pseudo-inverse of $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})$, and $\boldsymbol{e}$ is the shape control error. In Equation (29), the second term using a continuous saturation function is designed to eliminate the perturbations $\boldsymbol{\delta}$ in the measurement flow. The block diagram of the closed-loop system under our controller with online learning is demonstrated in Figure 3.

Under the proposed shape flow predictor, composite adaptive algorithm, and velocity controller, the stability of the closed-loop system is guaranteed and proved in Appendix A1, which gives rise to all the convergences of the shape control error $\boldsymbol{e}$, shape estimation error $\tilde{\boldsymbol{x}}$, and shape flow error $\dot{\tilde{\boldsymbol{x}}}$ to zeros as $t\to\infty$, that is,

$$\lim_{t\to\infty}\boldsymbol{e}=0,\quad \lim_{t\to\infty}\tilde{\boldsymbol{x}}=0,\quad \lim_{t\to\infty}\dot{\tilde{\boldsymbol{x}}}=0 \tag{30}$$

together with the convergence of estimated NN weights $\widehat{\boldsymbol{W}}_i$, $i\in\{1,2,\ldots,m\}$, and guaranteed NNs learning of the unknown model, if the predictor parameters α_x, β_x, and controller parameter k_s are chosen such that

$$k_s\geq b_{\delta 1},\quad \beta_x\geq b_{\delta 1}+\tfrac{b_{\delta 2}}{\alpha_x} \tag{31}$$

where $b_{\delta 1}$ and $b_{\delta 2}$ are positive constants in Equation (12).

Experimental Setup

To evaluate the proposed control algorithm, a comprehensive experimental study has been performed. We introduce the setup here, including general shape features, two platforms, FBG sensing system, and initialization.

General shape features

Two 3-D shape features, including point's position and bending/twist angles (called BTA), are defined, which can be composed to globally describe the shape of continuum robot. Point's position refers directly to the Cartesian position $\boldsymbol{r}_i$, $i\in\{1,2,\ldots,l\}$, selected on the continuum robot, and they can be utilized to construct the following features. BTA consist of bending angle κ and twist angle φ, which describe the deflection and torsion of one robot's section, respectively. Bending angle $\kappa\in\mathbb{R}$, for $0^\circ<\kappa\leq 180^\circ$, is a scalar representing the angle between two intersecting lines constructed from three reference points $\boldsymbol{r}_i$, $i\in\{1,2,3\}$, as shown in Figure 4A by

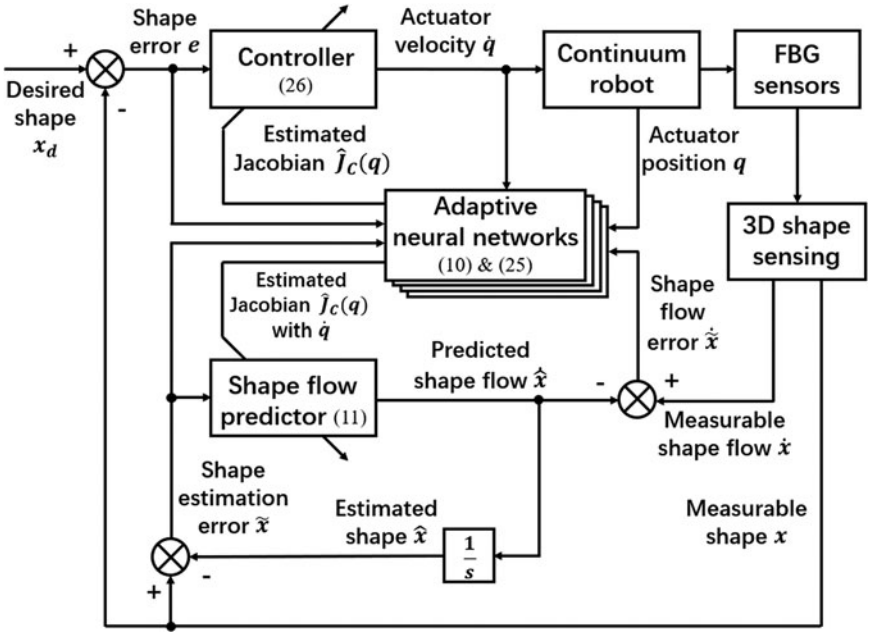


FIG. 3. Block diagram of the closed-loop system under the proposed control algorithm.

$$\kappa = \arccos\left(\frac{(\boldsymbol{r}_1 - \boldsymbol{r}_2) \cdot (\boldsymbol{r}_3 - \boldsymbol{r}_2)}{||\boldsymbol{r}_1 - \boldsymbol{r}_2||\,||\boldsymbol{r}_3 - \boldsymbol{r}_2||}\right) \tag{32}$$

Twist angle $\varphi \in \mathbb{R}$, for $-180^\circ < \varphi \leq 180^\circ$, describes the rotation about the x-axis of the base frame as shown in Figure 4B given by

$$\varphi = \text{atan2}(z_e,\ y_e) \tag{33}$$

where $\boldsymbol{r}_e = [\,x_e \quad y_e \quad z_e\,]^\mathsf{T} \in \mathbb{R}^3$ is the section's endpoint.

Robotic-assisted colonoscope system

We integrated a FBG-equipped robotic-assisted colonoscope system (called RACS here) previously reported,[19] of which the distal continuum mechanism with a 120 mm working length was used to perform the experiments, as shown in Figure 5A. A dial drive module was designed to control the deflection motions of the continuum mechanism with two DOFs (i.e., $n = 2$). We embedded a multicore FBG fiber in the working channel for proprioception, where the fiber tip was attached to the colonoscope tip. To globally describe the shape of the continuum part, two features are defined, including (1) two points' positions and (2) BTA. Two points' positions $\boldsymbol{x} = [\,\boldsymbol{r}_{e1}^\mathsf{T} \quad \boldsymbol{r}_{e2}^\mathsf{T}\,]^\mathsf{T} \in \mathbb{R}^6$ is constructed by stacking the Cartesian positions of distal endpoint $\boldsymbol{r}_{e1} \in \mathbb{R}^3$ and middle point $\boldsymbol{r}_{e2} \in \mathbb{R}^3$, and BTA $\boldsymbol{x} = [\,\kappa \quad \varphi\,]^\mathsf{T} \in \mathbb{R}^2$ consists of the bending and twist angles as introduced above.

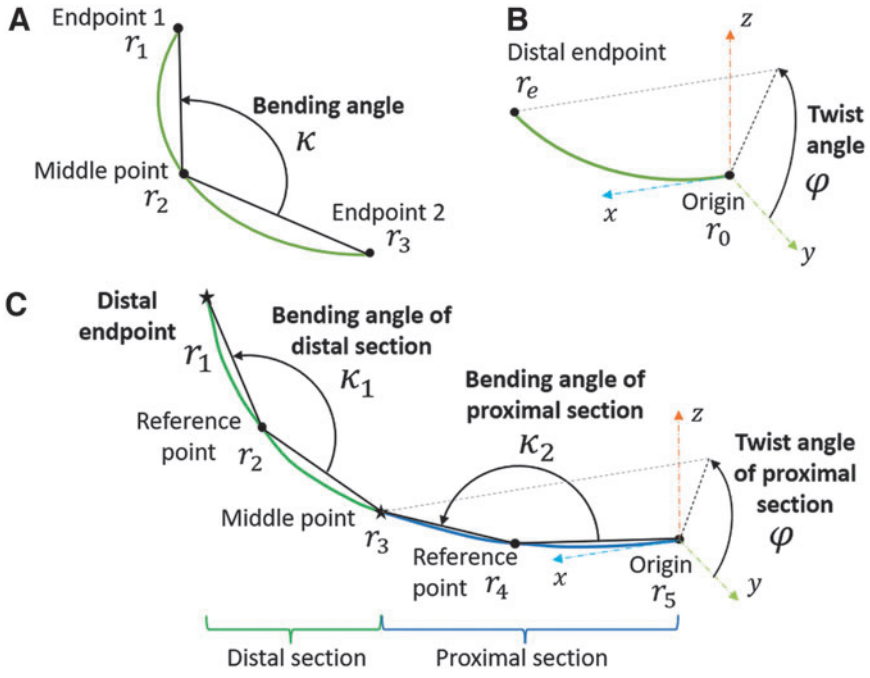


FIG. 4. 3-D shape features: **(A)** bending angle, **(B)** twist angle, and **(C)** DEP-BTA for multisection continuum manipulator. DEP-BTA, distal endpoint position with bending/twist angles.

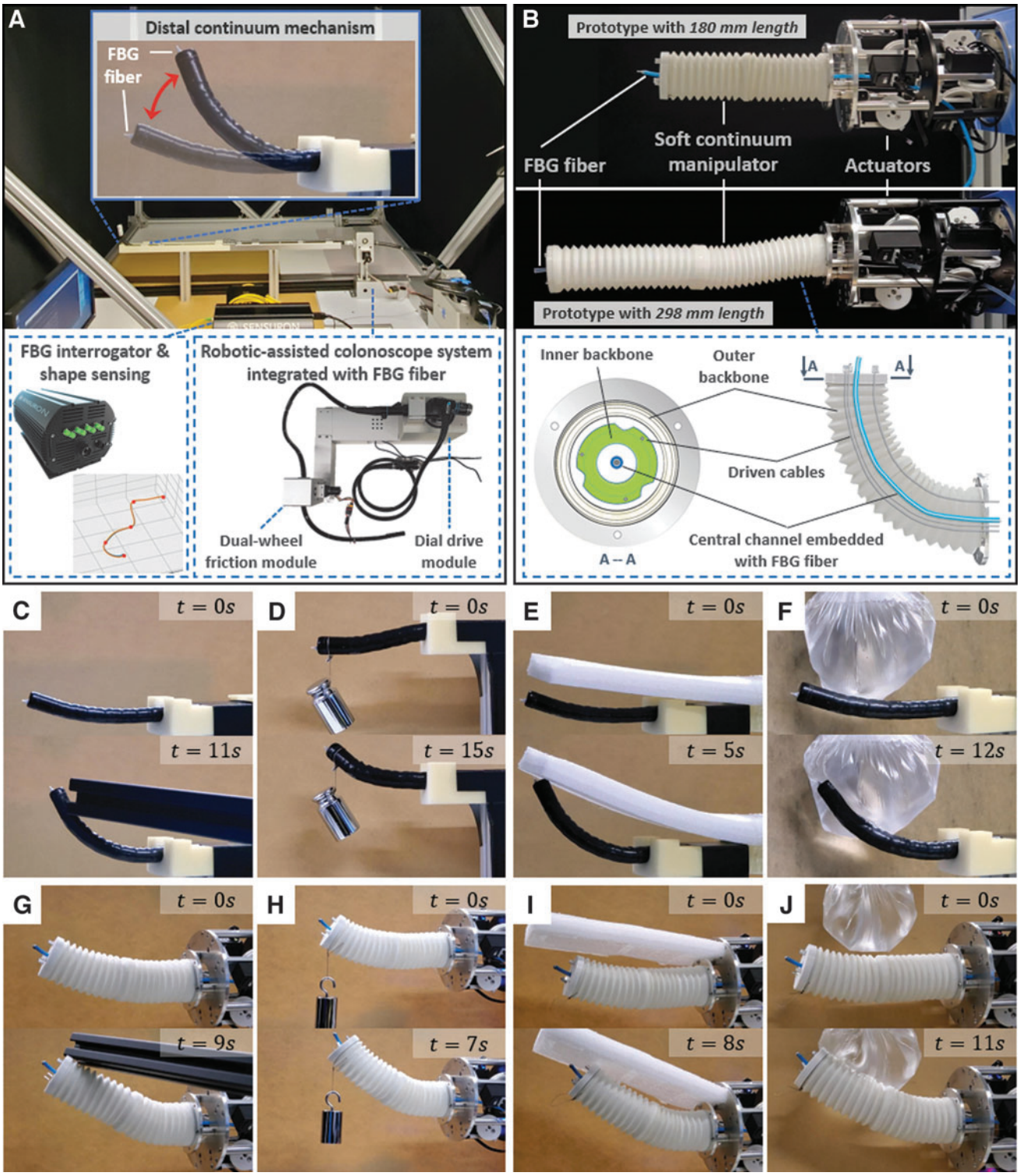


FIG. 5. Experimental platforms integrated with FBG fibers: **(A)** RACS; **(B)** multisection extensible SCM. Task setups using two platforms in unstructured environments with **(C)/(G)** a rigid rod, **(D)** 100 g/(H) 200 g payload, **(E)/(I)** elastic foam, and **(F)/(J)** a soft water bag. FBG, fiber Bragg grating; RACS, robotic-assisted colonoscope system; SCM, soft continuum manipulator.

Multisection extensible soft continuum manipulator

A soft continuum manipulator (called SCM here) with a 180 mm working length, two extensible sections, and six DOFs in total (i.e., $n = 6$) has been integrated and deployed for the experiments as shown in Figure 5B. Its extensible soft section, including outer and inner backbones fabricated by soft materials (creality thermoplastic polyurethane) and capable of omnidirectional deflections under variable stiffness/length, was previously used as a wrist component of a

robotic platform for OP swab sampling.[61,62] The deflection motions of each soft section are driven by three cables and actuated by servo motors. Filled with particles in the inner chamber of the soft section, its stiffness/length can be adjusted by pulling/pushing three cables together.[63] The FBG fiber was embedded in the central channel as shown in Figure 5B, where the fiber tip was fixed with the robot endpoint.

Besides two points' positions $\boldsymbol{x} \in \mathbb{R}^6$ as a type of shape feature, another shape description called distal endpoint position with bending/twist angles (DEP-BTA) $\boldsymbol{x} \in \mathbb{R}^6$, as shown in Figure 4C, is defined by stacking the Cartesian position of distal endpoint $\boldsymbol{r}_1 \in \mathbb{R}^3$ and three angles $\boldsymbol{\xi} \in \mathbb{R}^3$ (i.e., two bending angles of distal section κ_1 and proximal section κ_2, and one twist angle of proximal section φ) as

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{r}_1^{\mathsf{T}} & \boldsymbol{\xi}^{\mathsf{T}} \end{bmatrix}^{\mathsf{T}}, \quad \boldsymbol{\xi} = \begin{bmatrix} \kappa_1 & \kappa_2 & \varphi \end{bmatrix}^{\mathsf{T}} \tag{34}$$

where $\boldsymbol{r}_1$ can locate the desired distal endpoint and determine the total length of soft robot, κ_1 and κ_2 are utilized to describe the deflections of two sections, and φ can determine the direction of the intersection point of two sections.

FBG shape sensing system

To acquire the 3-D shapes of the continuum and soft robots, two four-core all-grating FBG fibers (FBGS International) with low reflective draw tower gratings at 1547 nm wavelength were separately integrated into the inner channels of two robots as shown in Figure 5, where the fiber tips were glued with the robot tips. The fiber has an outer diameter of 190 μm, including one central core, three outer cores positioned at 120° with 37 μm distance to the center, as well as Ormocer T coating. A fan-out box (FBGS International) with E2000/APC connection wires out four cores of the FBG fiber into individual single-mode fibers. Then, we used an interrogator (RTS125+; Sensuron) to simultaneously decode strains from 45 and 55 FBG sets of two robots, respectively, with both 30 Hz acquisition frequency and 3.3 mm spatial resolution.

Robust and accurate shape sensing results can be calculated and recorded from the measurement software using a new version of previous algorithm[19] with 25 Hz sensing frequency. The frequency of the overall closed-loop system under the proposed is 20 Hz, which is regarded as comparable with those in the related works.[14,21]

Initialization

The parameters adopted in the study were tuned based on the preliminary experiments providing the desired performance with convergences of shape control and estimation errors to zeros. In the experiments using RACS, the parameters were set as follows: $\alpha_x = 0.3$, $\beta_x = 0.04$, $k_e = 0.01$, $k_x = 0.01$, $k_r = 0.2$, $\boldsymbol{\Gamma}_W^{-1} = 0.1 \times \boldsymbol{I}_{18m \times 18m}$, $k_c = 0.32$, and $k_s = 0.04$. For online learning, we used nine neurons for each NN (i.e., $k = 9$) and RBFs as their activation functions. The centers $\boldsymbol{\mu}_{ij} \in \mathbb{R}^2$ of the i-th NN, $i \in \{1, 2\}$, were initialized by randomly sorting the columns of the chosen basis $\boldsymbol{\mu}_{0j} \in \mathbb{R}^2$, $j \in \{1, 2, \ldots, 9\}$, and so were the corresponding widths σ_{ij} from σ_{0j}, which aims to prevent the rows of the combined Jacobian from being linearly dependent. This can consequently avoid the Jacobian ill-conditioned or even singular, eliminating overspeed motions. For SCM, the parameters were set as: $\alpha_x = 0.6$, $\beta_x = 0.12$, $k_e = 0.01$, $k_x = 0.01$, $k_r = 0.2$, $\boldsymbol{\Gamma}_W^{-1} = \boldsymbol{I}_{78m \times 78m}$, $k_c = 0.02$, and $k_s = 0.01$. Thirteen neurons were utilized for each NN (i.e., $k = 13$), and the center and width values were initialized in the same way as they were when using RACS.

Given that there was no requirement of *a priori* identification for the robot's model, we initialized the estimated shape by zeros, that is, $\hat{\boldsymbol{x}} = \boldsymbol{0}_{m \times 1}$. To enable fast learning convergence and improve the transient performance of the proposed algorithm, a normalized initialization method[64] was utilized to initialize the NN weights. Then, we conducted a few iterations of continuous slow motions in a small region around the initial configuration in free environments,[58] which can speed up the convergence during shape servoing.

Experimental Results

Experimental validations (29 experiments in total) were performed to evaluate the performance of the proposed algorithm, including shape servoing tasks using two continuum robots in free and unstructured environments, validation for effectiveness of NNs online learning, validation on a longer soft continuum robot, validation with unreachable desired shape, comparison experiments, and case studies in medical contexts. The results, including Cartesian path of the robot's distal endpoint (CPDE), as well as two norms of shape errors, that is, those of shape control errors $\| \boldsymbol{e} \|$, shape estimation errors $\| \tilde{\boldsymbol{x}} \|$, and shape flow errors $\| \dot{\tilde{\boldsymbol{x}}} \|$, are recorded and presented.

In all figures showing CPDE and shape errors, the left plots show the resulting CPDE. Except special notes, the top-right images demonstrate the shape control errors, where Cartesian position errors $\Delta \boldsymbol{r}_{e1}, \Delta \boldsymbol{r}_{e2} \in \mathbb{R}^3$ of endpoint $\boldsymbol{r}_{e1}$ and middle point $\boldsymbol{r}_{e2}$ are presented for the errors described by two points. For the control errors described by BTA (using RACS), bending and twist errors $\Delta\kappa$, $\Delta\varphi \in \mathbb{R}$ are presented, and position error of distal endpoint $\Delta \boldsymbol{r}_1 \in \mathbb{R}^3$ with bending/twist error $\Delta\boldsymbol{\xi} \in \mathbb{R}^3$ is shown for those described by DEP-BTA (using SCM). The bottom-right plots indicate the shape estimation error $\tilde{\boldsymbol{x}}$ and shape flow error $\dot{\tilde{\boldsymbol{x}}}$ by blue solid curve and red dash-dotted curve, respectively. We attach Supplementary Videos S1 and S2 emphasizing the exterior views of continuum robots and simultaneous FBG-shape convergences.

Shape servoing tasks in free environments

We first conducted the shape control tasks in free space using RACS and SCM, respectively, and the resulting CPDE and shape errors are depicted in Figure 6, where two features were used to describe the shape of each robot. In the left plots of Figure 6A and B using RACS, distal endpoints can approach the desired positions with both shape features, although endpoint positioning was not processed when described by BTA. The right images indicate the asymptotic minimization of shape control and estimated errors regardless of different shape descriptions, showing that the 3-D shape of continuum mechanism can automatically converge to the desired one.

Given that the length of SCM can be adjusted, the arc lengths of its initial and desired shapes are different in all experiments, where the desired endpoint position is outside the initial workspace. The asymptotic convergences of shape errors in Figure 6C and D show that the robot's shapes

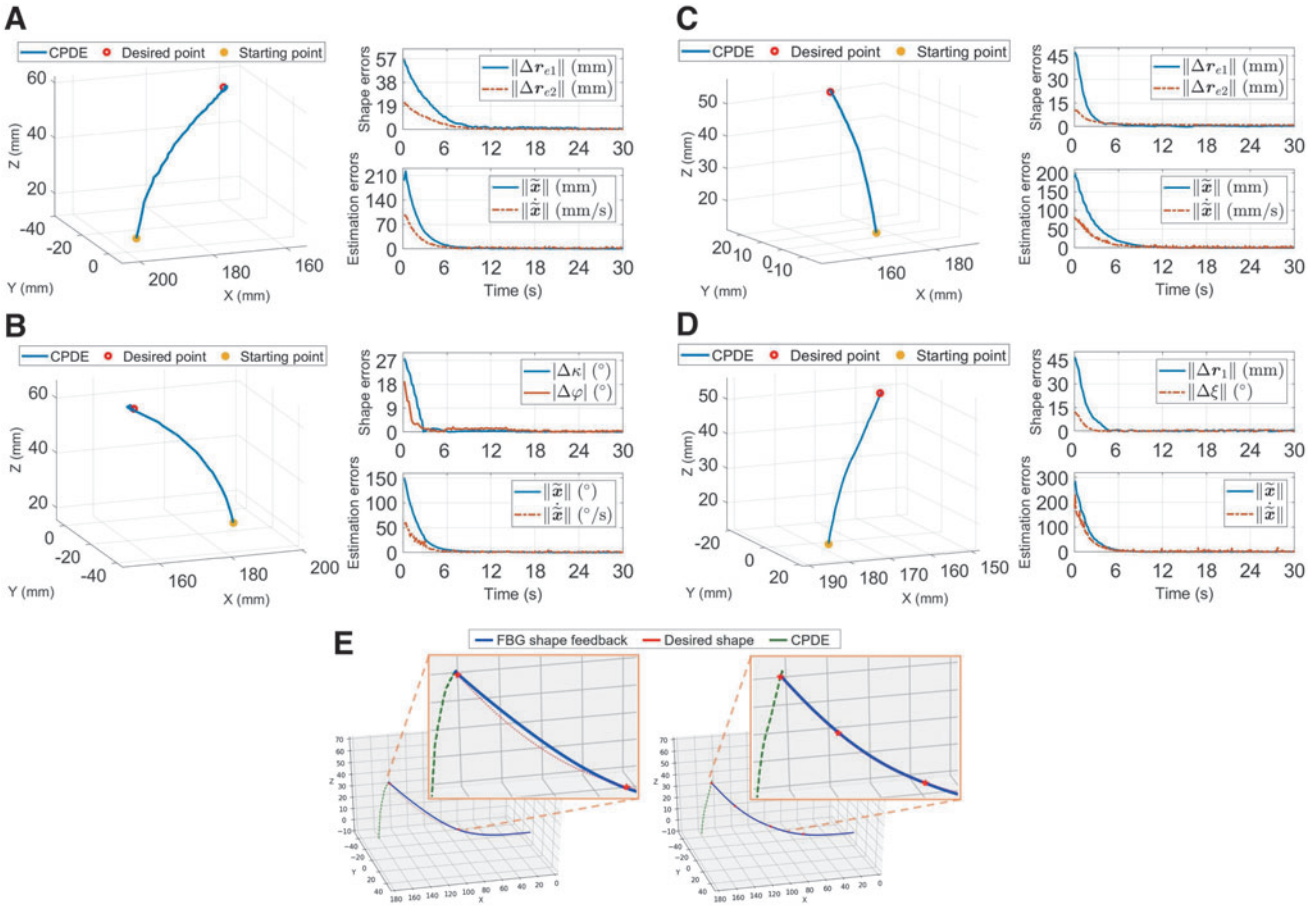


FIG. 6. CPDE (*left*) and shape errors (*right*) of experiments using RACS in free environments, with **(A)** two points and **(B)** BTA as shape features. Those using SCM with **(C)** two points and **(D)** DEP-BTA as shape features, and **(E)** final shapes described by two points (*left*) and DEP-BTA (*right*). CPDE, Cartesian path of the robot's distal endpoint.

described by two features are controlled to the desired configurations. Notice that the final shape described by two points in Figure 6E is different from the desired one, where the distal section is nearly stretched to a straight line, while that described by DEP-BTA incorporating endpoint positioning and two section's deflections imposes a desired shape.

Shape servoing tasks in unstructured environments

These experiments were performed using two continuum robots in unstructured environments as shown in Figure 5C–J, including the tasks with a rigid rod, payloads, foam, and a soft water bag. The results are presented in Figure 7, where two features were utilized to describe the robot shapes, and Supplementary Video S1 provides those using two points. Note that the initial and desired lengths of SCM are different in all experiments, during which the robot has to be extended for the desired shape.

Figure 7A and E indicates the first experimental results in collisions with the rigid rod, where we performed three random collisions to the distal parts of colonoscope and soft manipulator, respectively, using the rod as unexpected external disturbances as shown in Figure 5C and G. Notice that CPDE with obvious perturbations in the left images is plotted by gradient colors. We can find the sudden increases of the shape errors in the right figures, where the pink lines represent the time instants encountering the collisions. It is worth observing that the errors can keep converging after each collision, which is contributed by the robustness of shape flow predictor and controller, as well as the adaptability of NNs online learning despite intense perturbations.

We then evaluated the algorithm by applying two masses of 100 and 200 g weights, respectively, as extra unknown payloads on the tips of two robots as depicted in Figure 5D and H. The initial shapes suffer unexpected and obvious deformation compared with others in Figure 5. The results in Figure 7B and F illustrate that CPDE can approach the desired point, and the shape control and estimation errors are asymptotically minimized to zeros in the presence of unknown payload.

The third experiments were performed in collisions with an elastic foam as shown in Figure 5E and I, which mimic interacting with tissues during surgery, and the robot model during the interaction was different from that initially updated in free space. The left plots of Figure 7C and G depict that CPDE outputs a smooth path in free space, and it approaches the desired point by online approximation to the interaction model after the collision. Such performance is also illustrated in the right images, where the shape errors asymptotically converge to zeros.

In the fourth experiments, a soft water bag of 340 g weight hangs over the robots as illustrated in Figure 5F and J. The robot has to lift the bag regarded as a soft tissue and move to

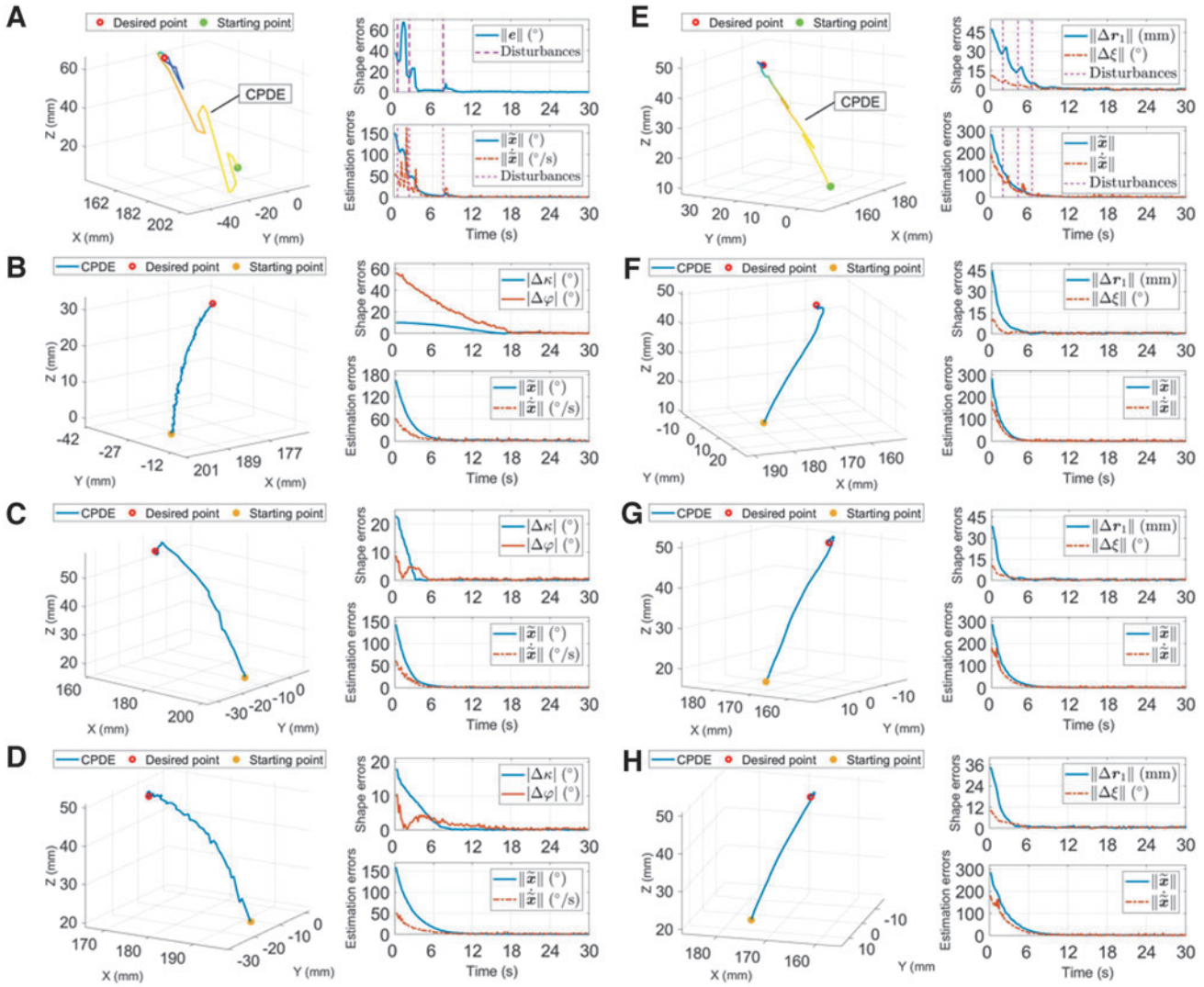


FIG. 7. CPDE (*left*) and shape errors (*right*) of experiments in unstructured environments, including the tests using RACS with **(A)** a rigid rod, **(B)** 100 g payload, **(C)** elastic foam, and **(D)** a soft water bag, and those using SCM with **(E)** a rigid rod, **(F)** 200 g payload, **(G)** elastic foam, and **(H)** a soft water bag.

the desired configuration, and it is concerned difficult to obtain *a priori* accurate model of the robot in this scenario considering multiple factors that can affect the robot's behavior (e.g., payloads and frictions). The results are shown in Figure 7D and H, where the shape errors decrease rapidly in free space and then converge to zeros during the interaction, which is contributed by the fast adaptation to unknown disturbances under the proposed controller. These results demonstrate that our algorithm can online learn the time-varying behavior of continuum robot, adaptively conform to unstructured environments, and simultaneously control the robot approaching the desired shape.

Validations on longer soft continuum robot

We performed three experiments using an SCM newly designed with two longer sections, the length of which was increased to 298 mm as shown in Figure 5B, and 90 FBG sets in the fiber were accordingly utilized for 3-D shape sensing and feedback. Due to the variable length of the soft manipulator, its initial and desired lengths are different. The first experiment was conducted using the longer SCM with a C-shape configuration, whose result can be found in Supplementary Video S1. S-shape configurations in 3-D space of the SCM[57] were evaluated in the second and third experiments as shown in Figure 8A and D, respectively, and presented in Supplementary Video S2.

The initial and final shapes of these two experiments are accordingly presented by blue curves in Figure 8B and E, where the red curves represent the desired shapes, and the green ones are the CPDE. Figure 8C and F demonstrates CPDE and asymptotic convergences of shape errors described by two points in these two experiments. These results show that the proposed algorithm is generic and feasible for longer continuum robots with both C-shape and S-shape configurations in 3-D space. The validation on another longer robot can refer to the afterward phantom experiment using RACS.

Effectiveness of NN online learning

To prove that the proposed algorithm is able to learn the combined Jacobian, five shape servoing tasks using RACS

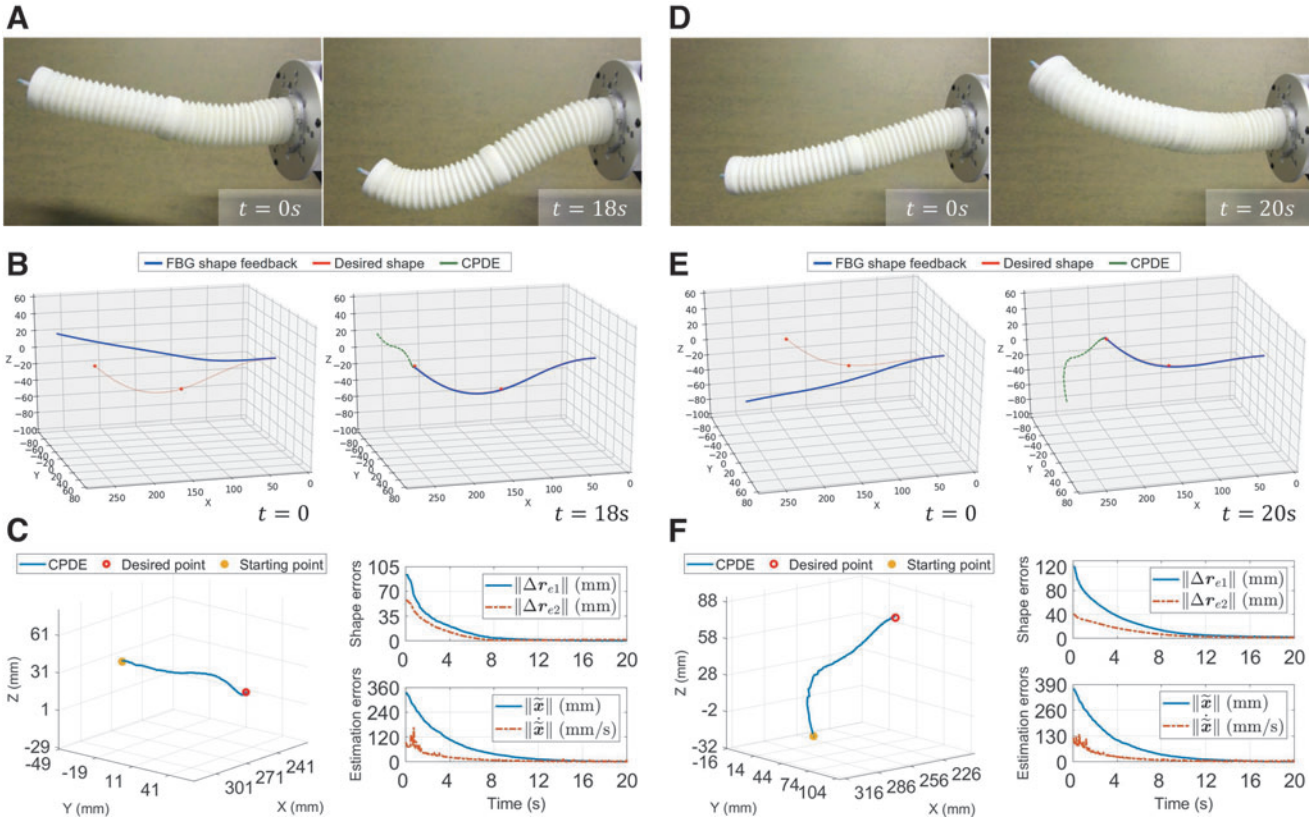


FIG. 8. Validations on longer SCM in two S-shape configurations with **(A)/(D)** task setups and snapshots, **(B)/(E)** initial (*left*) and final (*right*) robot shapes, and **(C)/(F)** corresponding CPDE (*left*) and shape errors (*right*).

were performed, and each task was repeated four times. After the first completion of each task, the afterward experiments were repeated and started by utilizing the weights updated from the previous completion. Note that the shape was controlled back to the same initial configuration after each completion, the processes of which are indicated by the white gaps in the top-right plot of Figure 9A, while those of the repeated experiments are marked and presented in the shade.

From the average convergence time (in terms of repeated times) of five tasks in Table 1 and the asymptotic convergences of shape errors in Figure 9A, it can be noticed that the latter three convergences are faster than the first one (by around 28.25%) during each task and they take a similar amount of time. Meanwhile, the distal endpoint has similar paths in the latter few convergences as shown in the left image. These results indicate that the proposed shape flow predictor together with the composite adaptation law has updated the weight matrices to steady values after the first attempt, and the learned Jacobian in the afterward repeated experiments only depends on the robot configuration.

Validation with unreachable desired shape

An experiment using RACS was conducted to validate the proposed method with an unreachable desired shape, where the desired position of distal endpoint is outside its workspace as shown in Figure 9C. CPDE and shape errors are depicted in Figure 9B, from which we can observe a steady-state error in the endpoint position error $\| \Delta \boldsymbol{r}_{e1} \|$ since its desired position is not feasible. Although the distal endpoint cannot be controlled to the desired place, the right plot of Figure 9C demonstrates that the shape of continuum mechanism can still approach the desired one using the proposed controller.

Comparison experiments

We conducted five tasks to compare the performance of the proposed approach with two model-based methods using a fixed model, a constant curvature model, and an iterative learning-based model, respectively, in terms of CPDE, shape control errors $\| \boldsymbol{e} \|$, and estimation errors $\| \tilde{\boldsymbol{x}} \|$. By deploying RACS with BTA as shape feature, the first two experiments were compared with a controller using a constant combined Jacobian $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})$ estimated around the starting point in free space, the purpose of which is to compare with a method that only utilizes a fixed model.[58] Their results can be found in Supplementary Video S1, where the fixed-model method demonstrates oscillation even divergence, while ours shows asymptotic convergences and robustness against unknown conditions.

The next two (third and fourth) experiments were performed by comparing with a controller using a constant curvature model,[1] where the robot shape was described by two points. The first test was processed by injecting the same unknown uncertainties into the actuator command input. Figure 9D demonstrates that although the results using the model-based method without modeling the uncertainties converge faster around the starting point than ours, its shape is not controlled to the desired one but oscillates in a region after 5.6 s. Our algorithm initially costs a few more iterations

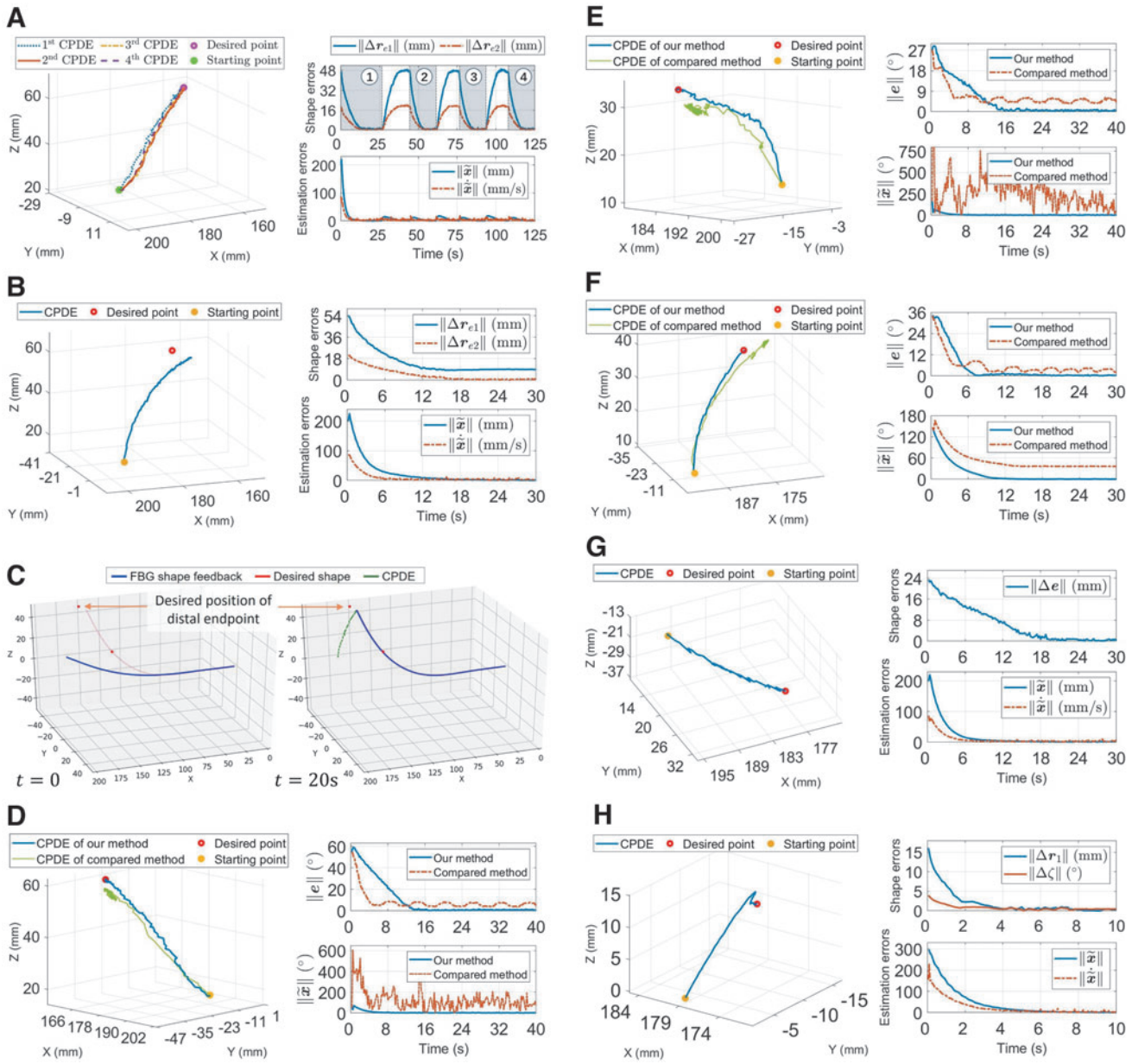


FIG. 9. CPDE (*left*) and shape errors (*right*) of validations. **(A)** effectiveness of NN online learning; **(B)** task with unreachable desired shape and **(C)** corresponding initial (*left*) and final (*right*) shapes; comparison results with model-based method using constant curvature with **(D)** internal uncertainties and **(E)** external disturbances in addition; **(F)** comparison results with iterative learning-based control method; **(G)** phantom experiment—shape control for robotic colonoscopy; **(H)** phantom experiment-OP swab sampling. OP, oropharyngeal.

for adaption to the time-varying model with unknown uncertainties, while its errors show asymptotic minimization to zeros at 11.2 s. In addition to the actuation uncertainties, we conducted the second test in a constrained environment with the soft water bag as depicted in Figure 5F. In Figure 9E, the shape control error using the model-based method does not show regulation but oscillations around a large value after the collision, while ours show the asymptotic convergences to zeros, which can be also indicated from the resulting CPDE.

Table 1. Average Convergence Time (s) of Experiments to Validate the Effectiveness of Neural Networks Online Learning

Repeat times	*1*	*2*	*3*	*4*
Average time (s)	15.33	11.00	11.40	10.87

We performed the last (fifth) task by comparing a method by iterative learning control[49] that allows model parameters learning of the soft robotic system in real time. Two methods were quantitatively compared according to the metrics in terms of CPDE, shape control errors $\| \boldsymbol{e} \|$, and shape estimation errors $\| \tilde{\boldsymbol{x}} \|$, which are presented in Figure 9F and in Supplementary Video S2 as well. In the left plot, the blue and green curves represent the CPDE using our method and

compared one, respectively, where CPDE controlled by the compared method oscillates in a local region near the desired point, while ours converges to the target. These can be also indicated from the results in the right plots.

Although the shape control error using the compared method shows faster convergence during the initial 5.5 s, showing its well model learning capability in the local region, it converges to a steady-state error with oscillations due to the unknown and time-varying conditions, and so does its estimation error. It is difficult for a priori knowledge of the robot model to be accurate when considering uncertainties and perturbations; from the results with these unexpected conditions, we can confirm that our learning-based controller can online update the unknown model of the continuum robot and robustly drive its shape toward the desired configuration.

Case studies in medical contexts

Two case studies were performed, including shape control for robotic colonoscopy and OP swab sampling. We deployed RACS with 400 mm length inside a colonoscopy phantom (Kyoto Kagaku M40) for automatic shape control of its distal continuum mechanism. The setup is shown in Figure 10A, where the distal part at the corner has to bend downwards to acquire a better endoscopic view for insertion guidance. Figure 10B presents the snapshots of endoscopic view controlled to a feasible one for continuing insertion. Note that RACS has a soft passive part with 280 mm length inside the phantom, which was not fixed and could collide with the phantom frequently resulting in unknown disturbances to the system. Hence, the results behave like some oscillations in Figure 9G, while they show convergences to the desired shape in the end.

The setup of OP sampling is illustrated in Figure 10C, where a swab is attached to the distal endpoint of SCM and it has to interact with the target region in a nasopharyngeal model (KJR/7208) for successful sampling. The swab is deformed by the interaction as shown in Figure 10D and E, during which the robot model is regarded as different from the initial one. Figure 9H shows that the endpoint position error $\| \Delta \boldsymbol{r}_1 \|$ behaves as slower convergence after the collision with the target, yet the algorithm adaptively keeps the error converging to zero with approaching the desired configuration. The results of phantom experiments support the claim that our controller can perform robust compliance to the unstructured environments, online learning of the time-varying model, and simultaneously servo-control the 3-D shape of continuum robot in these credible contexts.

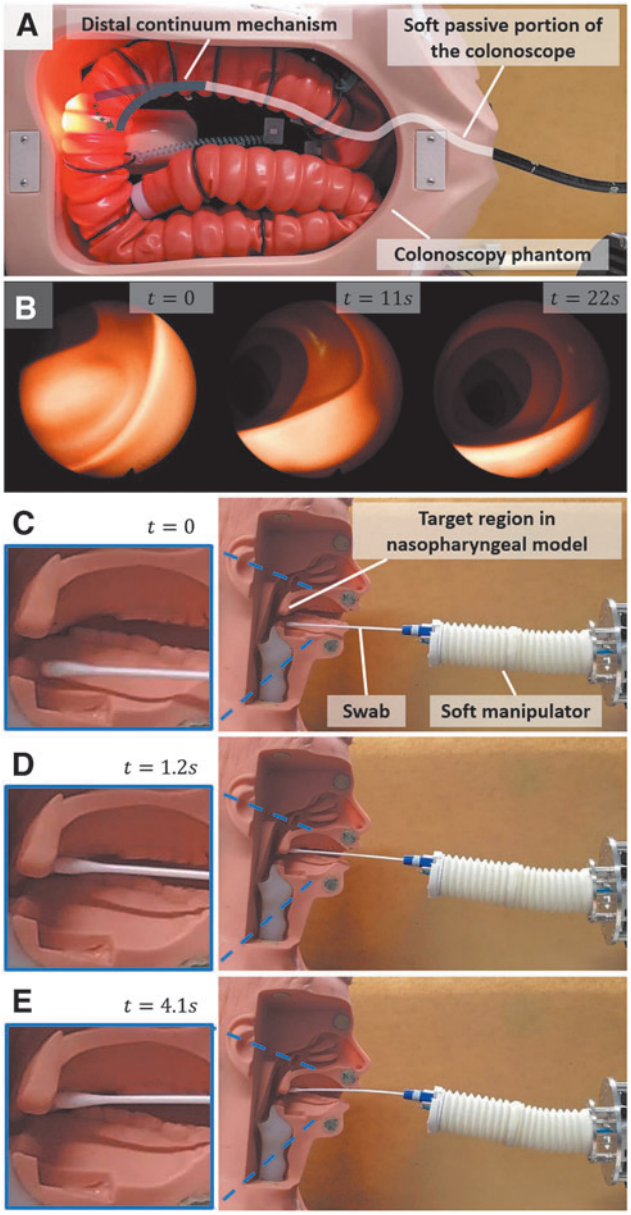


FIG. 10. Task setups and snapshots of phantom experiments. Shape control for robotic colonoscopy: **(A)** setup and **(B)** endoscopic views. OP swab sampling: **(C)** initial setup, **(D)** interaction with target, and **(E)** shape control completion.

Discussion

Apart from CPDE and shape errors presented above, the rank of the matrix $\widehat{\boldsymbol{J}}_c(\boldsymbol{q})\widehat{\boldsymbol{J}}_c^{+}(\boldsymbol{q}) \in \mathbb{R}^{m \times m}$ was monitored to prevent its reduction weakening the robot's manipulability. The control input $\dot{\boldsymbol{q}}$ was recorded as well and did not reach the threshold, thus avoiding fast motion of the robot. From all results using different continuum robots, we can observe that our control method can adaptively control the 3-D shape of continuum robot without any priori model in various unstructured environments. As for SCM featured with variable length, DEP-BTA composed of endpoint positioning, as well as bending/twisting of each soft section, is better to describe its shape for control tasks, rather than the feature of two points which may bring unexpected extension/compression to the soft sections.

The limitation of the approach is that the deployed shape sensing does not consider the variable arc length of SCM, although we can approximate and control its time-varying length by positioning its distal endpoint. We will enable an accurate length estimation using FBG sensors to cope with this issue. Besides, the proposed controller is generic that has the possibility of working for other robotic platforms with more soft sections or those with parallel configurations,[23,29] which will be investigated in our future works.

Conclusion

In this work, we propose a novel and generic data-driven approach for adaptive 3-D shape control of continuum and soft robots using FBG shape feedback in unstructured environments. An enhanced multicore FBG-based method with an accurate shape sensing algorithm is used to obtain the robot shape without external sensing units; thus, it can be utilized in MIS with occlusions of sight-of-view. The proposed algorithm enables robust model learning and simultaneous servo-control of the robot imposing the desired shape without any priori knowledge, thereby facilitating its applications with highly nonlinear uncertainties and unknown interactions. A learning-based shape flow predictor and a composite adaptation law are introduced for asymptotic regulations of shape estimation errors and online update of learning parameters. The stability of the close-loop control system has been theoretically proved by Lyapunov theory.

To evaluate the performance of the proposed method, detailed experiments have been conducted using two different continuum robots, including RACS and SCM, both equipped with multicore FBG fibers for 3-D shape sensing. It is worth noting that SCM is featured with variable length and stiffness. The results that consist of the converging endpoint paths and asymptotic regulations of shape control and estimation errors to zeros demonstrate the versatility, adaptability, and superiority of our method in various unstructured environments and phantom experiments against unexpected disturbances, which facilitate its practical deployments to medical continuum and soft robots.

Although we evaluate the performance of our method on the soft robot with desired configurations in C-shapes and S-shapes, those described in more complex shapes[55] will be investigated using a more flexible robot in the future. Since the variable length of SCM would affect the performance, a key issue is to acquire the accurate sensing length of the robot.[24] For this purpose, an assisting device and methodology using FBG sensors will be designed. Moreover, due to the versatility of the proposed controller, it can be extended for more practical tasks, for example, soft tissue manipulation[58] and eye-in-hand visual servoing.[16,43,44] By utilizing FBG data, the method will also be introduced in energy-based planning for underactuated control of continuum and soft robots within efficient and safe frameworks.

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Supplementary Material

Supplementary Video S1
Supplementary Video S2

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Appendix

Appendix A1 Stability Analysis of the Closed-Loop Control System

Define a Lyapunov candidate function as

$$\mathcal{V}=\frac{1}{2}k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e}+\frac{1}{2}k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}\tilde{\boldsymbol{x}}+\frac{1}{2}\widetilde{\boldsymbol{W}}^{\mathsf{T}}\Gamma_W\widetilde{\boldsymbol{W}}+k_r\mathcal{R} \tag{35}$$

with an auxiliary function $\mathcal{R}(t)$ defined as

$$\mathcal{R}(t)=\beta_x\sum_{i=1}^{m}|\tilde{\boldsymbol{x}}_i(0)|-\tilde{\boldsymbol{x}}^{\mathsf{T}}(0)\boldsymbol{\delta}(0)-\mathcal{H}(t)+\mathcal{H}(0) \tag{36}$$

where $i=1,2,\cdots,m$ represents the i-th element of $\tilde{\boldsymbol{x}}$, and the auxiliary function $\mathcal{H}(t)$[54] is generated by

$$\dot{\mathcal{H}}=\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}})) \tag{37}$$

The time derivative of $\mathcal{R}(t)$ is derived as

$$\dot{\mathcal{R}}=-\dot{\mathcal{H}}=-\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}})) \tag{38}$$

and $\mathcal{R}(t)$ is positive semidefinite if the sufficient condition

$$\beta_x\geq b_{\delta 1}+\frac{b_{\delta 2}}{\alpha_x} \tag{39}$$

in Equation (31) is satisfied, which will be proved in Appendix A2.

Differentiating $\mathcal{V}$ in Equation (35) and substituting Equations (8), (26), (28), and (38) results in

$$\begin{aligned}\dot{\mathcal{V}}=&k_e\boldsymbol{e}^{\mathsf{T}}(\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}}+\boldsymbol{d})+k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}\Theta^{\mathsf{T}}(\boldsymbol{q})\boldsymbol{Q}^{\mathsf{T}}(\dot{\boldsymbol{q}})\widetilde{\boldsymbol{W}}\\&+k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}(\boldsymbol{\delta}-\alpha_x\tilde{\boldsymbol{x}}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))\\&-\widetilde{\boldsymbol{W}}^{\mathsf{T}}(k_x\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(q)\tilde{\boldsymbol{x}}+k_e\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{e}\\&+k_r\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{r}_x)-k_r\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))\\=&-k_x\alpha_x\tilde{\boldsymbol{x}}^{\mathsf{T}}\tilde{\boldsymbol{x}}+k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))+k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}}\\&+k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{d}-k_e\widetilde{\boldsymbol{W}}^{\mathsf{T}}\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{e}\\&-k_r\widetilde{\boldsymbol{W}}^{\mathsf{T}}\boldsymbol{Q}(\dot{\boldsymbol{q}})\Theta(\boldsymbol{q})\boldsymbol{r}_x-k_r\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))\end{aligned} \tag{40}$$

Substituting Equations (25) and (27), we have

$$\begin{aligned}\dot{\mathcal{V}}=&-k_x\alpha_x\tilde{\boldsymbol{x}}^{\mathsf{T}}\tilde{\boldsymbol{x}}+k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))+k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}}\\&+k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{d}-k_e\boldsymbol{e}^{\mathsf{T}}\left(\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}}-\widehat{\boldsymbol{J}}_c(\boldsymbol{q})\dot{\boldsymbol{q}}-\boldsymbol{E}_J\dot{\boldsymbol{q}}\right)\\&-k_r\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{r}_x-\boldsymbol{\delta}+\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))-k_r\boldsymbol{r}_x^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))\\=&-k_x\alpha_x\tilde{\boldsymbol{x}}^{\mathsf{T}}\tilde{\boldsymbol{x}}+k_x\tilde{\boldsymbol{x}}^{\mathsf{T}}(\boldsymbol{\delta}-\beta_x\mathrm{sat}(\tilde{\boldsymbol{x}}))+k_e\boldsymbol{e}^{\mathsf{T}}\widehat{\boldsymbol{J}}_c(\boldsymbol{q})\dot{\boldsymbol{q}}\\&+k_e\boldsymbol{e}^{\mathsf{T}}\boldsymbol{\delta}-k_r\boldsymbol{r}_x^{\mathsf{T}}\boldsymbol{r}_x\end{aligned} \tag{41}$$

Using the controller in Equation (29) and taking the upper bound with Equation (12), we can further have

(Appendix continues →)

$$\begin{aligned}\dot{\mathcal{V}} =& -k_x\alpha_x\tilde{\boldsymbol{x}}^\mathsf{T}\tilde{\boldsymbol{x}} + k_x\tilde{\boldsymbol{x}}^\mathsf{T}(\boldsymbol{\delta} - \beta_x \text{sat}(\tilde{\boldsymbol{x}})) \\ &+ k_e\boldsymbol{e}^\mathsf{T}\boldsymbol{\delta} - k_e\boldsymbol{e}^\mathsf{T}(k_c\boldsymbol{e} + k_s\text{sat}(\boldsymbol{e})) - k_r\boldsymbol{r}_x^\mathsf{T}\boldsymbol{r}_x \\ \leq& -k_x\alpha_x\tilde{\boldsymbol{x}}^\mathsf{T}\tilde{\boldsymbol{x}} - k_x(\beta_x - b_{\delta 1})||\tilde{\boldsymbol{x}}|| - k_e k_c\boldsymbol{e}^\mathsf{T}\boldsymbol{e} \\ &- k_e(k_s - b_{\delta 1})||\boldsymbol{e}|| - k_r\boldsymbol{r}_x^\mathsf{T}\boldsymbol{r}_x\end{aligned} \tag{42}$$

From Equation (39), we can conclude that $-k_x(\beta_x - b_{\delta 1})||\tilde{\boldsymbol{x}}|| \leq 0$ since the following condition is satisfied

$$\beta_x \geq b_{\delta 1} + \tfrac{b_{\delta 2}}{\alpha_x} \geq b_{\delta 1} \tag{43}$$

If the controller parameter k_s is chosen sufficiently large such that $k_s \geq b_{\delta 1}$ in Equation (31) is satisfied, we have $-k_e(k_s - b_{\delta 1})||\boldsymbol{e}|| \leq 0$ and obtain that

$$\dot{\mathcal{V}} \leq -k_x\alpha_x\tilde{\boldsymbol{x}}^\mathsf{T}\tilde{\boldsymbol{x}} - k_e k_c\boldsymbol{e}^\mathsf{T}\boldsymbol{e} - k_r\boldsymbol{r}_x^\mathsf{T}\boldsymbol{r}_x \leq 0 \tag{44}$$

Since $\mathcal{V} > 0$ and $\dot{\mathcal{V}} \leq 0$, V are bounded, which directly implies that the shape control error $\boldsymbol{e}$, shape estimation error $\tilde{\boldsymbol{x}}$, and estimation error of NN weights $\widetilde{\overline{\boldsymbol{W}}}$ are all bounded. The boundedness of $\dot{\boldsymbol{q}}$ and that of activation functions $\boldsymbol{\theta}_i(\boldsymbol{q})$, $i \in \{1, 2, \ldots, m\}$, as well as their derivatives w.r.t. the corresponding arguments,[53] guarantee the boundedness of Θ and $\dot{\Theta}$ from Equation (19). Because $\boldsymbol{Q}$ and $\boldsymbol{\delta}$ are bounded by Equations (20) and (12), respectively, their boundedness together with those of Θ, $\widetilde{\overline{\boldsymbol{W}}}$, $\tilde{\boldsymbol{x}}$, and sat$(\cdot)$ function ensure that the shape flow error $\dot{\tilde{\boldsymbol{x}}}$ is bounded from Equation (26). From Equation (27), the boundedness of $\tilde{\boldsymbol{x}}$ and $\dot{\tilde{\boldsymbol{x}}}$ implies the boundedness of the filtered shape estimation error $\boldsymbol{r}_x$. The proj$(\cdot)$ function in Equation (28) indicates the boundedness of $\widehat{\overline{\boldsymbol{W}}}$, $\dot{\widehat{\overline{\boldsymbol{W}}}}$, and $\dot{\widetilde{\overline{\boldsymbol{W}}}}$.

Since $\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$ are bounded, and so are $\boldsymbol{W}_i$, $\boldsymbol{\theta}_i(\boldsymbol{q})$, $i \in \{1, 2, \ldots, m\}$, and $\boldsymbol{E}_J$,[53] $\boldsymbol{J}_c(\boldsymbol{q})$ and $\dot{\boldsymbol{Q}}$ are bounded from Equations (9) and (20), respectively. From Equation (8), the boundedness of $\dot{\boldsymbol{q}}$, $\boldsymbol{J}_c(\boldsymbol{q})$, and $\boldsymbol{d}$ also indicates that the shape feature is varied continuously, that is, $\dot{\boldsymbol{x}}$ is bounded and so is $\dot{\boldsymbol{e}}$. Since Θ, $\dot{\Theta}$, $\boldsymbol{Q}$, $\dot{\boldsymbol{Q}}$, $\widetilde{\overline{\boldsymbol{W}}}$, $\dot{\widetilde{\overline{\boldsymbol{W}}}}$, $\dot{\boldsymbol{\delta}}$ from Equation (12), sat$(\cdot)$ function with its derivative w.r.t. $\tilde{\boldsymbol{x}}$ and $\dot{\tilde{\boldsymbol{x}}}$ are all bounded, we can conclude that $\dot{\boldsymbol{r}}_x$ is bounded from Equation (27). Therefore, $\boldsymbol{e}$, $\tilde{\boldsymbol{x}}$, and $\boldsymbol{r}_x$ are uniformly continuous. By Barbalat's lemma, we can prove that

$$\lim_{t\to\infty} \boldsymbol{e} = \boldsymbol{0}, \quad \lim_{t\to\infty} \boldsymbol{r}_x = \boldsymbol{0}, \quad \lim_{t\to\infty} \tilde{\boldsymbol{x}} = \boldsymbol{0} \tag{45}$$

As $t \to \infty$, since $\boldsymbol{r}_x \to \boldsymbol{0}$ and $\tilde{\boldsymbol{x}} \to \boldsymbol{0}$, we have $\dot{\tilde{\boldsymbol{x}}} \to \boldsymbol{0}$ following Equation (27). From Equation (26), we can prove the asymptotic minimization of the estimation error of NN weights $\widetilde{\overline{\boldsymbol{W}}}$. Consequently, the shape control error, shape estimation error, and shape flow error are all convergent to zeros, and the online learning of the combined Jacobian $\boldsymbol{J}_c(\boldsymbol{q})$ using our adaptive NN approximation is guaranteed.

Appendix A2 Proof of Positive Semi-Definiteness of $\boldsymbol{R}(t)$

Recall that the auxiliary function $\mathcal{H}(t)$ is generated by

$$\dot{\mathcal{H}} = \boldsymbol{r}_x^\mathsf{T}(\boldsymbol{\delta} - \beta_x \text{sat}(\tilde{\boldsymbol{x}})) \tag{46}$$

By integrating $\dot{\mathcal{H}}$ and substituting $\boldsymbol{r}_x = \dot{\tilde{\boldsymbol{x}}} + \alpha_x\tilde{\boldsymbol{x}}$ in Equation (27), it is obtained that

$$\begin{aligned}\mathcal{H}(t) =& \mathcal{H}(0) + \int_0^t \left(\dot{\tilde{\boldsymbol{x}}} + \alpha_x\tilde{\boldsymbol{x}}\right)^\mathsf{T}(\boldsymbol{\delta} - \beta_x\text{sat}(\tilde{\boldsymbol{x}}))d\tau \\ =& \mathcal{H}(0) + \int_0^t \dot{\tilde{\boldsymbol{x}}}^\mathsf{T}\boldsymbol{\delta}d\tau - \int_0^t \beta_x\dot{\tilde{\boldsymbol{x}}}^\mathsf{T}\text{sat}(\tilde{\boldsymbol{x}})d\tau \\ &+ \int_0^t \alpha_x\tilde{\boldsymbol{x}}^\mathsf{T}(\boldsymbol{\delta} - \beta_x\text{sat}(\tilde{\boldsymbol{x}}))d\tau \\ =& \mathcal{H}(0) + \tilde{\boldsymbol{x}}^\mathsf{T}\boldsymbol{\delta} - \tilde{\boldsymbol{x}}^\mathsf{T}(0)\boldsymbol{\delta}(0) - \int_0^t \tilde{\boldsymbol{x}}^\mathsf{T}\dot{\boldsymbol{\delta}}d\tau \\ &- \beta_x\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(t)| + \beta_x\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(0)| \\ &+ \int_0^t \alpha_x\tilde{\boldsymbol{x}}^\mathsf{T}(\boldsymbol{\delta} - \beta_x\text{sat}(\tilde{\boldsymbol{x}}))d\tau\end{aligned} \tag{47}$$

Since $\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(t)| \geq ||\tilde{\boldsymbol{x}}||$, taking the upper bound of above equation yields

$$\begin{aligned}\mathcal{H}(t) \leq& \mathcal{H}(0) + ||\tilde{\boldsymbol{x}}||\,||\boldsymbol{\delta}|| - \tilde{\boldsymbol{x}}^\mathsf{T}(0)\boldsymbol{\delta}(0) \\ &+ \int_0^t ||\tilde{\boldsymbol{x}}||\,\|\,\dot{\boldsymbol{\delta}}\,\|\,d\tau - \beta_x||\tilde{\boldsymbol{x}}|| + \beta_x\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(0)| \\ &+ \int_0^t \alpha_x||\tilde{\boldsymbol{x}}||||\boldsymbol{\delta} - \beta_x\text{sat}(\tilde{\boldsymbol{x}})||d\tau\end{aligned} \tag{48}$$

Substituting the upper bounds of $\boldsymbol{\delta}$ and $\dot{\boldsymbol{\delta}}$ from Equation (12), we can obtain that

$$\begin{aligned}\mathcal{H}(t) \leq& \mathcal{H}(0) - \tilde{\boldsymbol{x}}^\mathsf{T}(0)\boldsymbol{\delta}(0) + \beta_x\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(0)| \\ &- (\beta_x - b_{\delta 1})||\tilde{\boldsymbol{x}}|| \\ &- \int_0^t \alpha_x\left(\beta_x - b_{\delta 1} - \frac{b_{\delta 2}}{\alpha_x}\right)||\tilde{\boldsymbol{x}}||d\tau\end{aligned} \tag{49}$$

We can conclude that the last two terms on the right of the above inequality are both negative semidefinite if the condition

$$\beta_x \geq b_{\delta 1} + \frac{b_{\delta 2}}{\alpha_x} \tag{50}$$

if Equation (31) is satisfied, then we further have

$$\mathcal{H}(t) \leq \mathcal{H}(0) - \tilde{\boldsymbol{x}}^\mathsf{T}(0)\boldsymbol{\delta}(0) + \beta_x\sum_{i=1}^m |\tilde{\boldsymbol{x}}_i(0)| \tag{51}$$

and prove $\mathcal{R}(t) \geq 0$ in Equation (36).